Question 1.1.1

Spring-Mass-Damper: Analytical and RK4 Analysis

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Problem Statement

Question:
A spring-mass-damper system with a mass of 1 kg is found to have a damping ratio of 0.2 and a natural frequency of 5 rad/s. The damping of the system is given by (AE 2007):

Options:

a) 2 Ns/m
b) 2 N/s
c) 0.2 kg/s
d) 0.2 N/m

Analytical Derivation

Governing Equation: $m\ddot{x} + c\dot{x} + kx = 0$

We are given:

- $m = 1$ kg
- $\zeta = 0.2$
- $\omega_n = 5$ rad/s

The damping ratio is defined as $\zeta = \frac{c}{2m\omega_n}$. Rearranging for c:

$$c = 2 \cdot \zeta \cdot m \cdot \omega_n$$
$$c = 2 \cdot (0.2) \cdot (1) \cdot (5) = \mathbf{2}$$

Dimensional analysis of damping force ($F_d = cv$) confirms the units are N/(m/s) = **Ns/m**.

System Modeling for RK4

To simulate this numerically using the 4th-Order Runge-Kutta (RK4) method, we must convert the 2nd-order ODE into a system of two 1st-order ODEs.

Let $x_1 = x$ (position) and $x_2 = \dot{x}$ (velocity).

$$\dot{x}_1 = x_2$$
$$\dot{x}_2 = \ddot{x} = -\frac{c}{m}\dot{x} - \frac{k}{m}x = -\frac{k}{m}x_1 - \frac{c}{m}x_2$$

With $m = 1$, $c = 2$, and $k = m\omega_n^2 = 25$, our state-space vector $\mathbf{Y}$ and its derivative $\mathbf{F}$ are:

$$\mathbf{Y} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, \quad \dot{\mathbf{Y}} = \mathbf{F}(t, \mathbf{Y}) = \begin{bmatrix} x_2 \\ -25x_1 - 2x_2 \end{bmatrix}$$

The RK4 Algorithm

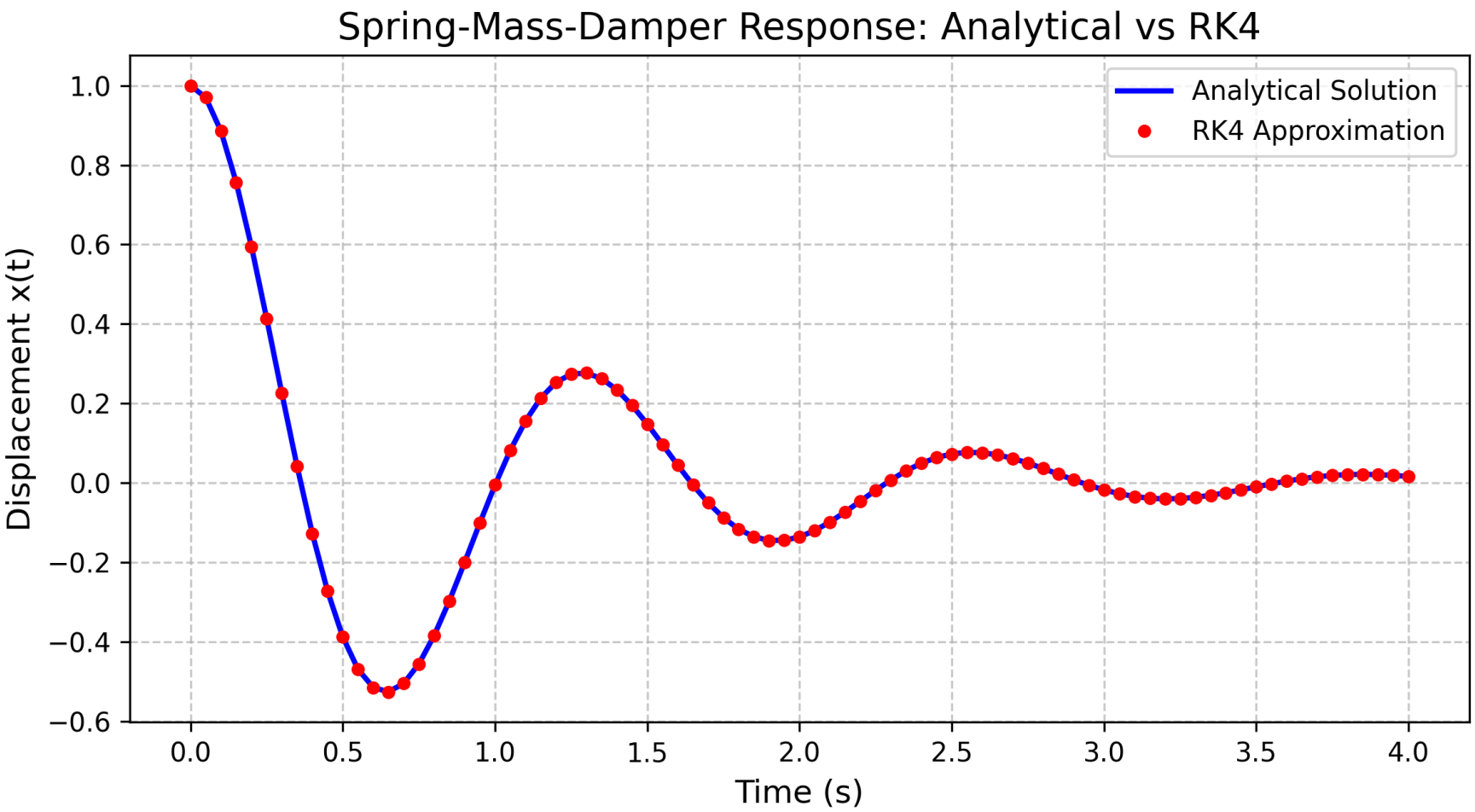
RK4 improves upon Euler by taking a weighted average of four slopes (k_1, k_2, k_3, k_4) across each time step h:

$$\mathbf{k}_1 = \mathbf{F}(t_i, \mathbf{Y}_i)$$
$$\mathbf{k}_2 = \mathbf{F}\left(t_i + \frac{h}{2}, \mathbf{Y}_i + \frac{h}{2}\mathbf{k}_1\right)$$
$$\mathbf{k}_3 = \mathbf{F}\left(t_i + \frac{h}{2}, \mathbf{Y}_i + \frac{h}{2}\mathbf{k}_2\right)$$
$$\mathbf{k}_4 = \mathbf{F}(t_i + h, \mathbf{Y}_i + h\mathbf{k}_3)$$

The next state is calculated as:

$$\mathbf{Y}_{i+1} = \mathbf{Y}_i + \frac{h}{6}(\mathbf{k}_1 + 2\mathbf{k}_2 + 2\mathbf{k}_3 + \mathbf{k}_4)$$

Verification: Analytical vs. RK4



Python Verification Script (RK4)

```python
import numpy as np
import matplotlib.pyplot as plt

# Parameters
m, c, k = 1.0, 2.0, 25.0
dt = 0.05
t = np.arange(0, 4 + dt, dt)

# State-space ODE function
def F(t, Y):
    x1, x2 = Y
    return np.array([x2, -(c/m)*x2 - (k/m)*x1])

# RK4 Implementation
Y = np.zeros((len(t), 2))
Y[0] = [1.0, 0.0] # Initial condition (x=1, v=0)

for i in range(len(t)-1):
    k1 = F(t[i], Y[i])
    k2 = F(t[i] + dt/2, Y[i] + dt*k1/2)
    k3 = F(t[i] + dt/2, Y[i] + dt*k2/2)
    k4 = F(t[i] + dt, Y[i] + dt*k3)
    Y[i+1] = Y[i] + (dt/6) * (k1 + 2*k2 + 2*k3 + k4)

# Plotting
zeta, wn = 0.2, 5.0
wd = wn * np.sqrt(1 - zeta**2)
x_ana = np.exp(-zeta*wn*t) * (np.cos(wd*t) + (zeta*wn/wd)*np.sin(wd*t))
```

Python Verification Script (RK4)

```
plt.figure(figsize=(8, 4))
plt.plot(t, x_ana, 'b-', label='Analytical', linewidth=2)
plt.plot(t, Y[:, 0], 'ro', markersize=4, label='RK4 Appx')
plt.title('Spring-Mass Response: Analytical vs RK4')
plt.xlabel('Time (s)'); plt.ylabel('Displacement')
plt.legend(); plt.grid(True, linestyle=':')
plt.tight_layout()
plt.savefig('rk4_spring_mass.png')
```

Conclusion

- Analytical calculation yields $c = 2$.
- Dimensional analysis verifies the unit is Ns/m.
- The RK4 numerical integration of the modeled state-space perfectly overlays the analytical solution, verifying our system parameters.

Final Answer

The correct option is **(a) 2 Ns/m**.